

Uses



Etched glass vase

These circuit breakers, containing a compound of fluorine and sulfur, cut electricity supply in an emergency.



Circuit breakers

These patterns form after the surface of the glass is burned by an acidic fluorine compound.

Ceramic pot



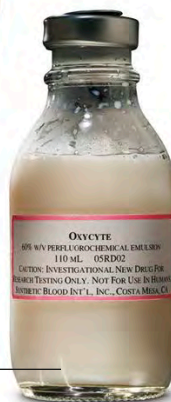
Fluorine-rich plastic clothing repels water.

The shiny coating on this ceramic dish is from a fluorine-containing glaze.



Waterproof clothing

Upon injection, this fluorine-rich liquid heals damaged tissues by carrying oxygen to them.



Oxycyte

PTFE was used by NASA to make spacesuits because it protected against heat.

This pan has a PTFE coating that resists heat.



Non-stick pan

Some toothpastes contain fluorides that harden tooth enamel.

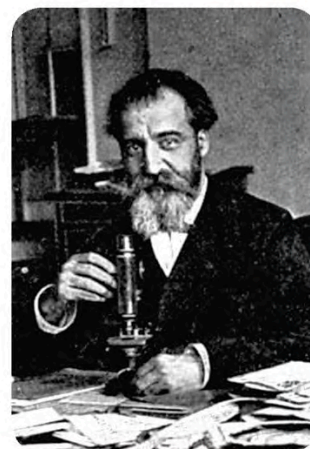


Toothpaste

HENRI MOISSAN



In the early 1800s, chemists in Europe realized that minerals such as fluorite contained an unknown element. However, it took another 70 years before the French chemist Henri Moissan extracted pure fluorine following a series of dangerous experiments that even poisoned him several times.



Some glazes used to coat ceramics contain fluorine minerals. When heated, these glazes release fluorine, which hardens the ceramic underneath. Another compound called polytetrafluoroethylene (PTFE) is commonly used to make **non-stick pans**: this material is

slippery and prevents food that has burned while cooking from sticking to the pan. Thin fibres made of PTFE are also used to make lightweight, **waterproof clothing**. One of the most common uses of fluorine compounds is in **toothpaste**: they toughen teeth against decay.